

# Francis Fan

📞 484-802-6820 | ✉ francis.fan@yale.edu | 🏠 Philadelphia, PA | 🔗 LinkedIn | 🐙 Github | 📁 Portfolio

## EDUCATION

### Yale University

Aug 2023 – May 2027

*Bachelor of Science in Computer Science*

*GPA: 3.96/4.00*

- **Activities:** Yale Club Swim (Captain), Yale Computer Society, Yale Undergraduate Capital Partners (Associate)
- **Courses:** Data Structures, Systems Programming and Computer Organization, Algorithms, Parallel Programming, Machine Learning, Compilers and Interpreters, Computational Intelligence for Games, Quantum Computing, Discrete Mathematics, Real Analysis, Linear Algebra, Software Engineering, Probability Theory, Theory of Statistics

## EXPERIENCE

### Ramen Inc.

May 2025 - Present

*Software Engineering Intern*

*Boston, MA*

- Engineered modular multi-step AI agents using MLflow, OpenAI APIs, and LangChain to deliver dynamic UI features such as user-adaptive components as well as allowing for users to access real-time diagnostics
- Tested and deployed AI agents with PostgreSQL, Kubernetes, and Docker, emphasizing stability and output coherence through automated testing of agent behavior against database state and system performance metrics

### Yale Computer Society

Sept 2024 - Present

*Lead Software Engineer for Yale Clubs*

*New Haven, CT*

- Produced club management webpage and other foundational features with secure student authorization using TypeScript, React.js, and CAS, reducing administrative workload and ensuring data protection for 6,000+ users
- Designed an AI-powered event verification tool using OpenAI APIs, Node.js, and MongoDB, integrated with JWT authentication, enabling secure and scalable event submissions beyond club activities, increasing user engagement

### Yale Department of Computer Science

Sept 2023 – Present

*Undergraduate Research Intern under Prof. Danny Rakita*

*New Haven, CT*

- **Project 1: Diffusion Policy for Manipulation** - Implemented diffusion policy, reinforcement learning (RL), and behavioral cloning script on models trained by GANs using ROS, PyTorch, and Python on dual robotic arms, optimizing viewpoints and object manipulation to achieve a 90% success rate across various movements
- **Project 2: Robotics Generative AI Reinforcement Learning Pipeline** - Used OpenRouter, PyTorch, and OpenAI Gymnasium to develop an agentic reward generation and task analysis pipeline to allow for autonomous RL capabilities and integrated ROS within a virtual environment with efficient publisher and subscriber nodes

## PROJECTS

### Brain Matter Data Analysis | SHAP, Random Forests, SVM, Multiple Linear Regression June 2022 - Aug 2023

- Utilized an interpretable model to quantify and confirm the cerebellum's importance in predicting risk tolerance and evaluated the performance of several machine learning techniques using 1000 runs of 10-fold cross-validation
- Published work using machine learning to assess the relationship between risk tolerance and brain gray matter volume as first author, and presented at the Organization for Human Brain Mapping (OHBM) Conference 2023

### Fit4Lyfe | MongoDB, Express, React, Node.js, Bcrypt, Mongoose, Recharts, GROQ, JWT Jan 2024 - July 2024

- Created full-stack web app to track workouts, visualize progress, and integrated AI insights for better training
- Deployed app on Render, serving 100+ users and 10,000+ data points through clean UI and optimized backend

### Avenue | Redis, React, Node.js, Serp API, Vercel KV, Firecrawl

Feb 2025 - Present

- Developed a price comparison web application and Chrome extension for finding related products at a lower price with optimized result caching using for fast, repeatable user queries with affiliate link APIs to monetize referrals
- Implemented AI-driven product matching using natural language embeddings and fuzzy search and Firecrawl for dynamic web crawling and HTML parsing to extract product metadata and pricing across multiple vendors

## ADDITIONAL

**Technical Skills:** Python (PyTorch, Scikit-learn), C/C++, Java, Javascript, ReactJS, Typescript, HTML, XML, SQL, R, CSS, Racket, x86-64 assembly, Git, Github, PostgreSQL, Jupyter Notebook, L<sup>A</sup>T<sub>E</sub>X(Overleaf/R Markdown), VS Code, Robot Operating System, Linux, Windows, Bash, Excel, React, Node.js, Figma, Jira, Jenkins, Vercel, Render, WordPress

**Preprints/Publications:** X. Sun, F. Fan, Y. Chen, D. Rakita. "A Comparative Study on State-Action Spaces for Learning Viewpoint Selection and Manipulation with Diffusion Policy." arXiv preprint, 2024. arXiv:2409.14615; Xiatao Sun, Shuo Yang, Yinxing Chen, Francis Fan, Yiyan (Edgar) Liang, Daniel Rakita. "Dynamic Rank Adjustment in Diffusion Policies for Efficient and Flexible Training" RSS, 2025